

Test of Significance for Single Proportion:-

Suppose a large random sample of size n has a sample proportion 'p' of members possessing a certain attribute (i.e., proportion of successes). To test the hypothesis that the proportion P in the population has a specified value P_0 .

Let us setup the Null hypothesis be $H_0: P = P_0$

then the alternative hypothesis is $H_1: P \neq P_0$ (or) $H_1: P > P_0$ (or) $H_1: P < P_0$

Since n is large, the sampling distribution of p is approximately normal.

$\therefore$ If H_0 is true, then the test statistic

$$Z = \frac{p - P_0}{S.E. \text{ of } p} \quad (\text{or}) \quad Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} \text{ is } \sim N(0, 1)$$

Problems:- ① In a sample of 1000 people in Karnataka 540 are rice eaters and the rest are wheat eaters. Can we assume that both rice and wheat are equally popular in this state at 1% level of significance?

Solution:- Sample size $n = 1000$

Let x : no. of rice eaters then $x = 540$,

p = Sample proportion of rice eaters $= \frac{x}{n} = \frac{540}{1000} = 0.54$

P = Population proportion of rice eaters $= \frac{1}{2} = 0.5$, Under

The assumption that both rice and wheat eaters are equally popular i.e. $H_0: P = 0.5$

Test Procedure:- (1) Null hypothesis: $H_0: P = 0.5$ i.e., Both rice and wheat eaters are equally popular in the state

(2) Alternative hypothesis $H_1: P \neq 0.5$ (Two-tailed test)

(3) $\alpha = 0.01$, Critical value Z_α at $\alpha = 0.01$ for two-tailed test

is $\boxed{Z_\alpha = 2.58}$

(4) Test statistic: $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.54 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{1000}}} = 2.532$,

$\boxed{|Z| = 2.532}$

(5) Conclusion: $|Z| = 2.532$, $Z_{\alpha} = 2.58$

$|Z| < Z_{\alpha}$: We Accept H_0 at 1% Level of Significance

$\therefore$ We Conclude that Both rice and wheat eaters are equally popular in the state of Karnataka.

(2) In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?

Solution:- Given. Sample size $n = 600$

Let X : number of smokers : ie $X = 325$

p = Sample proportion of smokers = $\frac{X}{n} = \frac{325}{600} = 0.5417$

P = Population proportion of smokers in the city = $\frac{1}{2} = 0.5$

Under the assumption that smokers and non smokers are equal in the city ie $H_0 : P = 0.5$; $Q = 1 - P = 0.5$

Test Procedure : (1) $H_0 : P = 0.5$

(2) $H_1 : P > 0.5$ (majority of men in this city are smokers)

We can use Right-tailed test.

(3) Level of Significance: $\alpha = 0.05$ (assume)

Critical value Z_{α} at $\alpha = 0.05$ for Right-tailed test is $Z_{\alpha} = 1.645$

(4) Test statistic: $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.5417 - 0.5}{\sqrt{\frac{(0.5)(0.5)}{600}}} = 2.04$

$\therefore$ calculated value of $Z = 2.04$

(5) Conclusion:- $|Z| = 2.04$, $Z_{\alpha} = 1.645$

$|Z| > Z_{\alpha}$: We Reject H_0

$\therefore$ We conclude that the majority of men in the city are smokers.

Problem (3):- 20 People were attacked by a disease and only 18 Survived. Will you reject the hypothesis that the Survival rate if attacked by this disease is 85% in favour of the hypothesis that is more at 5% level.

Solution:- Given, Sample size $n = 20$

X : number of Survived people ; $x = 18$

P = Proportion of Survived people $= \frac{x}{n} = \frac{18}{20} = 0.9$

$P = 0.85$, Under that assumption that the ~~hypothesis~~ Survival rate if attacked by disease is 85%.

$$Q = 1 - P = 1 - 0.85 = 0.15 \quad ; \quad \boxed{Q = 0.15}$$

Test Procedure:- (1) $H_0 : P = 0.85$
(2) $H_1 : P > 0.85$ (Right tailed test)
(3) $\alpha = 0.05$

critical value Z_α at $\alpha = 0.05$ for one tailed test is $Z_\alpha = 1.645$
ie $\boxed{Z_\alpha = 1.645}$

(4) Test Statistic:- $Z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.9 - 0.85}{\sqrt{\frac{(0.85)(0.15)}{20}}} = \frac{0.05}{0.08} = 0.625$

$$\boxed{|Z| = 0.625}$$

(5) Conclusion:- $|Z| = 0.625$, $Z_\alpha = 1.645$

$|Z| < Z_\alpha$: We Accept H_0

ie The proportion of the Survival rate is 0.85.

Practice Problems:-

- ① Experience had shown that 20% of a manufactured product is of the top quality. In one day's product of 400 articles only 50 are of top quality. Test the hypothesis at 0.05 level.
- ② A Social worker believes that fewer than 25% of the ~~cou~~ couples in a certain area have ever used any form of birth control. A random sample of 120 couples was contacted. Twenty of them said that they have used. Test the belief of the social worker at 0.05 level of significance
- ③ A manufacturer claims that only 4% of his products are defective. A random sample of 500 were taken among which 100 were defective. Test the hypothesis at 0.05 level.
- ④ A manufacturer claimed that at least 95% of the equipment which he supplied to a factory conformed to specifications. An examination of a sample of 200 pieces of equipments received 80 were faulty. Test the claim at 0.05 level.

Test for equality of two Proportions (or) Test for Significant Difference between two proportions

Let P_1 and P_2 be the proportions in two large random samples of sizes n_1 and n_2 drawn from two populations having proportions P_1 and P_2 . To test whether the two population proportions P_1, P_2 are equal.

The Null hypothesis $H_0: P_1 = P_2$

(The Alternative hypothesis $H_1: P_1 \neq P_2$

(let us assume that H_0 is true. then we have $P_1 = P_2 = P$ (say)

and the Sampling distribution of $P_1 - P_2$ is approximately normal with mean = 0 and $S.D = \sqrt{pq(\frac{1}{n_1} + \frac{1}{n_2})}$

Where $p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$ and $q = 1 - p$

Since the Sample Sizes n_1 and n_2 are large, the test statistic

$$Z = \frac{P_1 - P_2}{S.E. of (P_1 - P_2)} = \frac{P_1 - P_2}{\sqrt{pq(\frac{1}{n_1} + \frac{1}{n_2})}} \sim N(0,1) \quad \text{Where } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}, q = 1 - p$$

If the Sample proportions are not known, then we use

$$|Z| = \frac{|P_1 - P_2|}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$$

Problems:- ① In two large populations, there are 30% and 25% respectively of fair haired people. Is this difference likely to be hidden in Samples of 1200 and 900 respectively from the two populations.

Solution:- Given $n_1 = 1200$, $n_2 = 900$

P_1 = Proportion of fair haired people in the first population

$$\text{ie } P_1 = 30\% = \frac{30}{100} = 0.3$$

P_2 = Proportion of fair haired people in the Second population

$$\text{ie } P_2 = 25\% = \frac{25}{100} = 0.25$$

Test Procedure:- (1) $H_0: P_1 = P_2$ (Assume that the Sample proportions are equal) ie the difference in population proportions is likely to be hidden in Sampling

(2) Alternative hypothesis $H_1: P_1 \neq P_2$ (two tailed test)

(3) Level of Significance: $\alpha = 0.05$ (Assume)

Z_{α} at $\alpha = 0.05$ for two tailed test is $Z_{\alpha} = 1.96$

(4) Test statistic: $|Z| = \frac{|P_1 - P_2|}{\sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}}$

Where $Q_1 = 1 - P_1 = 1 - 0.3 = 0.7$
 $Q_2 = 1 - P_2 = 1 - 0.25 = 0.75$

$$\therefore Z = \frac{0.3 - 0.25}{\sqrt{\frac{(0.3)(0.7)}{1200} + \frac{(0.25)(0.75)}{900}}} = 2.55$$

$$\boxed{|Z| = 2.55}$$

(5) Conclusion:- $|Z| = 2.55$; $Z_\alpha = 1.96$

$|Z| > Z_\alpha$; We Reject the Null hypothesis H_0

$\therefore$ We Conclude that the Sample proportions are not equal.

thus We Conclude that the difference in population proportions is unlikely that the real difference will be hidden.

Problem(2):- on the basis of their total Scores, 200 Candidates of a Civil Service examination are divided into two groups, the upper 30% and the remaining 70%. Consider the first question of the examination. Among the first group, 40 had the Correct answer, whereas among the second group, 80 had the Correct answer. on the basis of these results, can one Conclude that the first question is not good at discriminating ability of the type being examined here?

Solution:- Given, 30% of 200 Candidates are upper group.

ie $n_1 = \frac{30}{100} \times 200 = 60$

$$\boxed{n_1 = 60}$$

70% of 200 Candidates are remaining group. ie $\frac{70}{100} \times 200 = 140$

ie $\boxed{n_2 = 140}$

$x_1 = 40$ (ie Among the first group had the correct answer)

$x_2 = 80$ (ie Among the second group had the correct answer)

$$\begin{aligned} \text{then } p_1 &= \frac{x_1}{n_1} = \frac{40}{60} = \frac{2}{3} = 0.667 \\ p_2 &= \frac{x_2}{n_2} = \frac{80}{140} = \frac{4}{7} = 0.571 \end{aligned} \quad \left\{ \begin{aligned} p &= \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{40 + 80}{60 + 140} = 0.6 \\ q &= 1 - p = 1 - 0.6 = 0.4 \end{aligned} \right.$$

Test Procedure: (1) Null hypothesis $H_0: p_1 = p_2$

(2) Alternative hypothesis $H_1: p_1 \neq p_2$

(3) level of significance $\alpha = 0.05$ (assumed)

(4) Test statistic:
$$Z = \frac{p_1 - p_2}{\sqrt{pq \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{0.667 - 0.571}{\sqrt{(0.6)(0.4) \left(\frac{1}{60} + \frac{1}{140} \right)}} = 1.27$$

$$|Z| = 1.27$$

(5) Conclusion: - $|Z| = 1.27$ $Z_\alpha = 1.96$

$|Z| < Z_\alpha$; we Accept H_0 at $\alpha = 5\%$

ie The first question is good enough in discriminating the ability of the students of both groups.

Problem (3): - In a City A, 20% of a random sample of 900 school boys had a certain slight physical defect. In another city B, 18.5% of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant at 0.05 level of significance.

Answer: - Given that, $n_1 = \text{first Sample Size} = 900$
 $n_2 = \text{Second Sample Size} = 1600$

$$x_1 = 20\% \times 900 = \frac{20}{100} \times 900 = 180$$

$$x_2 = 18.5\% \times 1600 = \frac{18.5}{100} \times 1600 = 296$$

$$\therefore p_1 = \frac{x_1}{n_1} = \frac{180}{900} = 0.2, \quad p_2 = \frac{x_2}{n_2} = \frac{296}{1600} = 0.185$$

$$\text{Thus } p = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 296}{900 + 1600} = \frac{476}{2500} = 0.19$$

$$\text{Now } q = 1 - p = 1 - 0.19 = 0.81$$

Test Procedure :- (1) Null hypothesis $H_0: p_1 = p_2$

There is no significance difference between population proportions.

(2) Alternative hypothesis $H_1: p_1 \neq p_2$ (two tailed test)

(3) Level of Significance $\alpha = 0.05$

Critical value z_α at $\alpha = 0.05$ for two tailed test is

$$\boxed{z_\alpha = 1.96}$$

$$(4) \text{ Test statistic: } z = \frac{p_1 - p_2}{\sqrt{pq\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{0.2 - 0.185}{\sqrt{(0.19)(0.81)\left(\frac{1}{900} + \frac{1}{1600}\right)}}$$

$$z = \frac{-0.015}{0.01634} = -0.918$$

$$\boxed{|z| = 0.918}$$

(5) Conclusion :- $|z| = 0.918, z_\alpha = 1.96$

$|z| < z_\alpha$ ie We Accept the Null hypothesis H_0 .

We conclude that there is no significance difference between the proportions

Practice Problems: (1) Random Samples of 400 men and 600 women were asked whether they would like to have a flyover near their residence. 200 men and 325 women were in favour of the proposal. Test the hypothesis that proportions of men and women in favour of the proposal are same, at 5% level?

(2) In a sample of 600 students of a certain college 400 are found to use ball pens. In another college, from a sample of 900 students 450 were found to use ball pens. Test whether the two colleges are significantly different with respect to habit of using ball pens.

(3) A Cigarette manufacturing firm claims that its brand A line of cigarette outsells its brand B by 8%. If it is found that 42 out of a sample of 200 smokers prefer brand A and 18 out of another sample of 100 smokers prefer brand B, Test whether the 8% difference is a valid claim.

Confidence interval (or) fiducial limits

The Computation of Confidence interval and Confidence limits is based on the Sampling distribution of a Statistic. The Sampling distribution of the Sample mean $\bar{x}$ of a random Sample of size n ($n \geq 30$) drawn from a population having mean μ and S.D. σ is approximately a normal distribution with mean μ and S.D. = S.E of $\bar{x} = \sigma/\sqrt{n}$

$\therefore z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ has a standard normal distribution with mean '0', S.D.=1

Confidence limits for population mean ' μ ' at 95% Confidence is given by

$$|z| \leq 1.96 \quad \text{ie} \quad \left| \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right| \leq 1.96$$

$$|\bar{x} - \mu| \leq 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

$$-1.96 \frac{\sigma}{\sqrt{n}} \leq \bar{x} - \mu \leq 1.96 \frac{\sigma}{\sqrt{n}}$$

$$1.96 \frac{\sigma}{\sqrt{n}} \geq -\bar{x} + \mu \geq -1.96 \frac{\sigma}{\sqrt{n}} \quad \left(\begin{array}{l} \text{multiplying with } - \text{ on} \\ \text{both sides} \end{array} \right)$$

$$\bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \geq \mu \geq \bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \quad \left(\text{Add } \bar{x} \text{ on both sides} \right)$$

$$\therefore \boxed{\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}}} \text{ are Confidence limits}$$

for population mean μ at 95% Confidence.

ie., $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ are Confidence limits for μ

$z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$ is called maximum error of Estimate. (E).

$$\therefore \boxed{E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}}$$

Similarly 99% Confidence limits for population mean ' μ ' is

$$\text{given by } \boxed{\bar{x} - 2.58 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 2.58 \frac{\sigma}{\sqrt{n}}}$$

Similarly 90% Confidence limits for population mean ' μ ' is

$$\text{given by } \boxed{\bar{x} - 1.645 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + 1.645 \frac{\sigma}{\sqrt{n}}}$$

Confidence limits for Population proportion 'P'

- (i) 95% Confidence limits are $p \pm 1.96 \sqrt{\frac{pq}{n}}$
- (ii) 99% " " " " $p \pm 2.58 \sqrt{\frac{pq}{n}}$
- (iii) 90% " " " " $p \pm 1.645 \sqrt{\frac{pq}{n}}$

Problem ① A random Sample of Size 100 is taken from a population with $\sigma = 5.1$. Given that the Sample mean is $\bar{x} = 21.6$. Construct a 95% Confidence interval for the population mean ' μ '.

Solution:- Given $\bar{x}$ = Sample mean = 21.6

n = Sample Size = 100

$\sigma = 5.1$
95% Confidence limits for population mean ' μ ' is $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$

$$21.6 \pm (1.96) \frac{5.1}{\sqrt{100}} \quad \text{i.e., } 21.6 - 0.9996 \leq \mu \leq 21.6 + 0.9996$$

$$\text{i.e., } 21.6 - 1 \leq \mu \leq 21.6 + 1 \quad (\text{approximately})$$

$$20.6 \leq \mu \leq 22.6$$

Hence (20.6, 22.6) is the Confidence interval for the population mean ' μ '

Problem ②:- The mean and standard deviation of a population are 11795 and 14054 respectively. If $n = 50$, find 95% Confidence limits and 99% Confidence limits for the mean.

Solution:- Given that $n = 50$, $\bar{x} = 11795$, $\sigma = 14054$

95% Confidence limit are $(\bar{x} - (1.96) \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + (1.96) \frac{\sigma}{\sqrt{n}})$

$$\text{i.e. } (11795 - \frac{(1.96)(14054)}{\sqrt{50}} \leq \mu \leq 11795 + \frac{(1.96)(14054)}{\sqrt{50}})$$

$$\text{i.e. } (11795 - 3895.6 \leq \mu \leq 11795 + 3895.6)$$

$$\text{i.e. } (7899.4 \leq \mu \leq 15690.6)$$

99% Confidence limits are $(\bar{x} - (2.58) \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + (2.58) \frac{\sigma}{\sqrt{n}})$

$$(11795 - 5733.1 \leq \mu \leq 11795 + 5733.1)$$

$$6061.9 \leq \mu \leq 17528.1$$

Problem(3): A random sample of 400 items is found to have mean 82 and S.D of 18. Find the maximum Error of estimate at 95% Confidence, and find the Confidence limits for the mean(μ) if $\bar{x} = 82$.

Solution: Given that $n = \text{Sample Size} = 400$
 $\bar{x} = \text{Sample mean} = 82$
 $S = \text{Sample S.D} = 18$

$$\text{Maximum error } E = Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \quad (\sigma \text{ is not known})$$

$$E = (1.96) \frac{S}{\sqrt{n}} = (1.96) \frac{18}{\sqrt{400}} \quad (95\% \text{ Confidence})$$

$$E = 1.764$$

95% Confidence limits for mean ' μ ' is given by

$$\bar{x} - E \leq \mu \leq \bar{x} + E$$

$$\text{ie } 82 - 1.764 \leq \mu \leq 82 + 1.764$$

$$\boxed{80.236 \leq \mu \leq 83.764}$$

Problem(4): - A random Sample of 500 pineapples was taken from a large Consignment and 65 were found to be bad. Find the percentage of bad pineapples in the Consignment.

Solution: - Given $n = \text{Sample Size} = 500$
 $X : \text{Bad pineapples}$

$$p = \text{Proportion of bad pineapples in the Sample} = \frac{x}{n} = \frac{65}{500} = 0.13$$

$$q = 1 - p = 1 - 0.13 = 0.87$$

We don't know the population proportion, to estimate population proportion by sample proportion then the limits for population proportion is given by $p \pm (1.96) \sqrt{\frac{pq}{n}}$ (95% Confidence)

$$\text{ie } p - (1.96) \sqrt{\frac{pq}{n}} \leq p \leq p + (1.96) \sqrt{\frac{pq}{n}}$$

$$0.13 - 0.0012 \leq p \leq 0.13 + 0.0012 \Rightarrow 0.1288 \leq p \leq 0.1312$$

$$0.1288 \leq p \leq 0.1312 \quad \therefore 12.88 \leq p \leq 13.12$$

Problem (5):- Among 900 people in a state 90 are found to be chapatti eaters. Construct 99% Confidence interval for the true proportion.

Solution:- Given $n = 900$

$x = \text{no. of chapatti eaters, i.e. } x = 90$

$$p = \frac{x}{n} = \frac{90}{900} = \frac{1}{10} = 0.1, q = 1 - p = 1 - 0.1 = 0.9$$

New P is not given $\therefore$ estimate P by using p

$$\sqrt{\frac{pq}{n}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{(0.1)(0.9)}{900}} = 0.01$$

99% Confidence interval for population proportion is given by

$$p - 2.58 \sqrt{\frac{pq}{n}} \leq P \leq p + 2.58 \sqrt{\frac{pq}{n}}$$

$$0.1 - (2.58) \sqrt{\frac{(0.1)(0.9)}{900}} \leq P \leq 0.1 + (2.58) \sqrt{\frac{(0.1)(0.9)}{900}}$$

$$0.1 - (2.58)(0.01) \leq P \leq 0.1 + (2.58)(0.01)$$

$$0.1 - 0.0258 \leq P \leq 0.1 + 0.0258$$

$$\boxed{0.0742 \leq P \leq 0.1258}$$

Problem (6):- A Sample of size 300 was taken whose variance is 225 and mean 54. Construct 95% Confidence interval for the mean.

Problem (7):- A Sample of size 100 was taken from a population S.D of Sample is 0.3. Find the maximum error with 99% Confidence.

Problem (8):- What is the size of the smallest sample required to estimate an unknown proportion to within a maximum error of 0.06 with at least 95% Confidence.
(Hint: P is not given. So we take $P = \frac{1}{2}$, thus $q = \frac{1}{2}$)

Problem (9):- If we can assert with 95% that the maximum error is 0.05 and $P = 0.2$. find the size of the sample.